Dr. Carlos Cifuentes San Román

Curriculum Vitæ
December 2024

1 Current position

- *Postdoctoral researcher* in *the* K-dwarfs Orbited By habitable Exoplanets (KOBE) project at the Centro de Astrobiología (CAB). European Space Astronomy Centre (ESAC) in Madrid, Spain.

A full list of previous positions is available in Appendix A.

2 Expertise and interests

- **Detection and characterisation of exoplanets** using the radial-velocity method.
- **Stellar characterisation of cool dwarfs**: Fundamental parameters and stellar multiplicity. Determination of spectral parameters in the infrared spectrum.
- **Unresolved multiplicity** based on the *Gaia* astrometric solution.
- **Presence and survival** of short-period planets in hierarchical triple systems.
- **Catalogue** building and maintainance.

3 Publication summary

- **100** publications, including: **53** refereed papers (published in 1st quartile of impact factor journals: *A&A*, *ApJ*, *Science*), **3** as a first author, **8** as a second, third, or *last* author, **32** VizieR catalogues, **1** circular.
- **16** *proceedings* of national and international meetings; **10** oral contributions, **14** poster contributions.
- **1876** citations; Hirsch h-index = **24** (n publications with $\geq n$ citations); Hirsch m-index = **3.0** ($m = h/(n_{2024} - n_{2018})$); i_{10}-index = **35** (number of publications with ≥ 10 citations).

More details are available in Appendices B and C. An up-to-date library can be found in ADS.

4 Observational experience

- **4** approved proposals as PI (**8** full nights, **4** half nights, and **5** service nights) and **6** approved proposals as co-PI (**20+** full nights).
- **13** nights on telescope as observer or co-observer at NOT, WHT, INT, 2.2-m CAHA, and **1** remote night co-observing with VLT.

More details are available in Appendix D.

5 Degrees

- *PhD* **Astrophysics**, UCM/CAB/MPIA, Madrid, 2023.
- *MSc* **Astrophysics**, UCM. Madrid, 2017.
- *MSc* **Renewable energies**, UAM, Madrid, 2014.
- *BSc/Lic.* **Physics**, USAL, Salamanca, 2013.

More details are available in Appendix E.

6 Skills

6.1 Computational

Beginner means *used at some point*; Operational means *extensive experience with*; Proficient means *day-to-day comfort.*

	Beginner	Operational	Proficient
Operating Systems		Linux, Windows	macOS
Programming	`C`, `C#`	Shell Script (Unix)	`Python`
Databases		`SQL (ADQL)`	
Collaboration & Version control		Git	
Mathematics & Statistics	MATLAB, SPSS	Mathematica, R	
Virtual Observatory	Splat, Stilts		TOPCAT, VOSA
Astronomy (Databases)			CDS[1]
Astronomy (Exoplanet analysis)	`exoplanet`	`juliet`, `exostriker`	
Astronomy (Spectra reduction)	IRAF, DS9		
Document preparation			LaTeX[2] , `#Markdown`
Web development & design	`jQuery`	`html, CSS, BS5`	
Image, audio & video edition		Adobe PS, BR	iMovie, Final Cut Pro

6.2 Languages

Español	Mother tongue
English	High level speaking, listening and reading (C1)
Deutsch	Some speaking, listening and reading ability (∼A1–A2)

[1] Centre de Données astronomiques de Strasbourg (CDS) includes ADS, Simbad, VizieR, and Aladin.

[2] Additionally, office suites such as Microsoft Office (Word, Powerpoint, Excel) and macOS iWork (Pages, Keynote, Numbers). Typing speed: 47–48 wpm / 235–240 cpm ($\geq$ 95% accuracy) – `@keybr`

7 Personal details

Full name	Cifuentes San Román, Carlos
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8 References

- Dr. Jorge Lillo-Box (CAB; jlillo@cab.inta-csic.es, jlillobox.com)
- Dr. José Antonio Caballero Hernández (CAB; caballero@cab.inta-csic.es, exoterrae.eu)
- Dr. Jorge Sanz Forcada (CAB; jsanz@cab.inta-csic.es)
- Dr. María Rosa Zapatero Osorio (CAB; mosorio@cab.inta-csic.es)
- Dr. A. Schweitzer (HS; andreas.schweitzer@hs.uni-hamburg.de)
- Dr. Trifon Trifonov (MPIA; trifonov@mpia.de, trifonov.in)
- Prof. Dr. David Montes Gutiérrez (UCM; dmontes@cab.inta-csic.es)
- Prof. Dr. Ángel Morales Sabio (CIEMAT; amorales@ciemat.es) → Recom. letter

CAB: Centro de Astrobiología; CIEMAT: Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas; HS: Hamburger Sternwarte; MPIA: Max-Planck-Institut für Astronomie; UCM: Universidad Complutense de Madrid.

A Professional experience

A.1 Research

- **Predoctoral Researcher** – Centro de Astrobiología (CAB), Madrid.
  Dr. José Antonio Caballero and Dr. Jorge Sanz Forcada. Jul 2018–Mar 2023 (**4.5 years**).
- **Predoctoral Internship** – Max-Planck-Institut für Astronomie (MPIA), Heidelberg.
  Dr. Trifon Trifonov. Sep–Dec 2021 (**3 months**).
- **Research Personnel in training** – Universidad complutense de Madrid (UCM), Madrid.
  Dr. José Antonio Caballero and Prof. David Montes. Nov 2016–Sep 2017 (**10 months**).
- **Research Personnel in training** – Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas (CIEMAT), Madrid.
  Dr. Ángel Morales Sabio and Dr. Nuria Germán Cordero. Jan–Jun 2014 (**5 months**).

A.2 Refereeing

- *Qeios*: 1

A.3 Teaching

- **Professor** – *Introduction to Astronomy* for science and non-science majors. Saint Louis University (SLU), Madrid. Jan–May 2024 (**1 semester**).
- **Professor** – *Differential Geometry*. Mathematics BSc degree at Universidad Internacional de Empresa (UNIE), Madrid. Jan–Jun 2024 (**1 semester**).

A.4 Private company

- **Consultant** – Indra Sistemas S.A. / Minsait, Smart Energy, Madrid. Nov 2014–Jul 2016 (**1.7 years**). I gained experience in:
  - Design, development, deployment, and monitoring if energy efficiency solutions for national and international projects.
  - Preparation of proposals and technical documentation, conduction of market studies, development of pricing models, and drafting of international bids (e.g., Programme H2020).
  - Presentation of reports, demos, and solutions for energy monitoring systems, utilizing communication protocols such `ZigBee`, `zWave`, `Modbus`.
  - Support and collaboration in project management duties in a highly interdisciplinary environment.

A.5 Grants and funding

Plan Nacional projects

- **PID2023-150468NB-I00**: "Formación estelar y planetaria: la diversidad y habitabilidad según JWST Y PLATO" (CAB-INTA). PIs: Dr. David Barrado Navascués and Dr. Nuria Huélamo Bautista.
- **PID2019-109522GB-C51**: "Enanas marrones y planetas aislados y alrededor de estrellas" (CAB-CSIC). PIs: Dr. María Rosa Zapatero Osorio and Dr. José A. Caballero Hernández.
- **AYA2016-79425-C3-2-P**: "Enanas marrones y planetas aislados y como compañeros de estrellas" (CAB-CSIC). Subprograma de Formación de Personal Investigador (FPI). Granted amount: 92 843 EUR; Administered budget: 6250 EUR. PIs: Dr. María Rosa Zapatero Osorio and Dr. Jorge Sanz Forcada.

Other national projects

- **MDM-2017-0737**: Unidad de Excelencia María de Maeztu "Evaluando la emergencia de vida como un fenómeno universal mediante la exploración planetaria" (CAB).

B Research

For publications that are not first-authored, the individual contribution is indicated in upper script, summarised as follows: (1) Writing, reviewing literature, formatting and editing; (2) Stellar characterisation and discussion of stellar features, including derivation of stellar parameters ($\mathcal{L}$, $\mathcal{M}$, $\mathcal{R}$; $T_{\rm eff}$); (3) Data collection, manipulation and analysis; (4) Plotting and visualisation elements; (5) RV time series analysis and planet characterisation, (6) Other (specified).

B.1 Selection of 1st-, 2nd-, or 3rd-authored publications

1. **C. Cifuentes**, et al. "The CARMENES input catalogue of M dwarfs. V. Luminosities, colours, and spectral energy distributions". *A&A*, 642, A115. (**135** citations)
2. **C. Cifuentes**, et al. "The CARMENES input catalogue of M dwarfs. IX. Multiplicity from close spectroscopic binaries to ultrawide systems". *Astronomy & Astrophysics*, in press.
3. A. Schweitzer, V. M. Passeger, **C. Cifuentes**[1,2,3], et al. "The CARMENES search for exoplanets around M dwarfs. Radii and masses of the target stars". *A&A* 625, A68. (**160** citations)
4. J. González-Payo, J. A. Caballero, **C. Cifuentes**[1,3], et al. "Characterisation of all multiple systems within 10 pc". *Astronomy & Astrophysics*, in prep.
5. H. Martínez-Rodríguez, J. A. Caballero, **C. Cifuentes**[1,2,3], et al. "Exomoons in the habitable zones of M dwarfs". *ApJ*, 887, 2, 261. (**33** citations)

B.2 Publications with peer review process

53. **C. Cifuentes**, J. A. Caballero, J. González-Payo, P. J. Amado, V. J. S. Béjar, et al. "The CARMENES input catalogue of M dwarfs. IX. Multiplicity from close spectroscopic binaries to ultra-wide systems". *Astronomy & Astrophysics*, in press.
52. O. Balsalobre-Ruza, J. Lillo-Box, A. M. Silva, et al. (incl. **C. Cifuentes**[4,5]). "KOBE-1: The first planetary system from the KOBE survey. Two planets likely in the sub-Neptune mass regime around a late K-dwarf". *Astronomy & Astrophysics*, in press.
51. H. L. Ruh, M. Zechmeister, A. Reiners, E. Nagel, Y. Shan, **C. Cifuentes**[1,2], et al. "The CARMENES search for exoplanets around M dwarfs The impact of rotation and magnetic fields on the radial velocity jitter in cool stars". *Astronomy & Astrophysics*, in press.
50. M. Cortés-Contreras, J. A. Caballero, D. Montes, C. Cardona-Guillén, V. J. S. Béjar, **C. Cifuentes**[1,2], et al. "CARMENES input catalogue of M dwarfs VIII. Kinematics in the solar neighbourhood". *Astronomy & Astrophysics*, in press.
49. J. González-Payo, J. A. Caballero, J. Gorgas, M. Cortés-Contreras, M. C. Gálvez-Ortiz, **C. Cifuentes**[1,2]. "Multiplicity of stars with planets in the solar neighbourhood". *Astronomy & Astrophysics*, 689, A302.
48. A. von Stauffenberg, T. Trifonov, A. Quirrenbach, S. Reffert, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. Revisiting the GJ 581 multi-planetary system with new Doppler measurements from CARMENES, HARPS, and HIRES". *Astronomy & Astrophysics*, 688, A112.
47. P. Mas-Buitrago, A. González-Marcos, E. Solano, V. M. Passegger, et al. (incl. **C. Cifuentes**[1,2]). "Using autoencoders and deep transfer learning to determine the stellar parameters of 286 CARMENES M dwarfs". *Astronomy & Astrophysics*, 687, A205.
46. E. Goffo, P. Chaturvedi, F. Murgas, G. Morello, et al. (incl. **C. Cifuentes**[1,2]). "TOI-4438 b: a transiting mini-Neptune amenable to atmospheric characterization". *Astronomy & Astrophysics*, 685, A147.
45. S. Dreizler, R. Luque, I. Ribas, V. Koseleva, et al. (incl. **C. Cifuentes**[1,2]). "Teegarden's Star revisited: A nearby planetary system with at least three planets". *Astronomy & Astrophysics*, 685, A117.
44. F. Llorente de Andrés, R. de la Reza, R., P. Cruz, et al. (incl. **C. Cifuentes**[1,2]). "The evolution of lithium in FGK dwarf stars. Influence of planets and Galactic migration". *Astronomy & Astrophysics*, 684, A28.

43. Y. Shan, D. Revilla, S. L. Skrzypinski, S. Dreizler, et al. (incl. **C. Cifuentes**[1,2]). "CARMENES input catalog of M dwarfs: VII. New rotation periods for the survey stars and their correlations with stellar activity". *Astronomy & Astrophysics*, 684, A9.

42. M. Mallorquín, E. Goffo, E. Pallé, N. Lodieu, et al. (incl. **C. Cifuentes**[1,2]). "TOI-1801 b: a temperate mini-Neptune around a young M0.5 dwarf". *Astronomy & Astrophysics*, 680, A76.

41. P. Gorrini, J. Kemmer, S. Dreizler, R. Burn, T. Hirano, et al. (incl. **C. Cifuentes**[1,2]). "Planetary companions orbiting the M dwarfs GJ 724 and GJ 3988 A CARMENES and IRD collaboration". *Astronomy & Astrophysics*, 680, A28.

40. E. Pallé, J. Orell-Miquel, M. Brady, J. Bean, A. P. Hatzes, et al. (incl. **C. Cifuentes**[1,2]). "GJ 806 (TOI-4481): A bright nearby multi-planetary system with a transiting hot, low-density super-Earth". *Astronomy & Astrophysics*, 678, A80.

39. J. Orell-Miquel, M. Lampón, M López-Puertas, M. Mallorquín, et al. (incl. **C. Cifuentes**[1,2]). "Confirmation of an He I evaporating atmosphere around the 650-Myr-old sub-Neptune HD 235088 b (TOI-1430 b) with CARMENES". *Astronomy & Astrophysics*, 677, A56.

38. E. González-Álvarez, M. R. Zapatero Osorio, V. J. S. Béjar, **C. Cifuentes**[1,2], et al. "Two sub-Neptunes around the M dwarf TOI-1470". *Astronomy & Astrophysics*, 675, A177.

37. E. González-Álvarez, J. Kemmer, P. Chaturvedi, J. A. Caballero, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. A sub-Neptunian mass planet in the habitable zone of HN Lib". *Astronomy & Astrophysics*, 675, A141.

36. A. Bello-García,V. M.Passegger, J. Ordieres-Meré, A. Schweitzer, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs–A deep transfer learning method to determine $T_{\rm eff}$ and [M/H] of target stars". *Astronomy & Astrophysics*, 673, A105.

35. R. de la Reza, C. Chavero, S. Roca-Fábrega, F. Llorente de Andrés, P. Cruz, and **C. Cifuentes**[1,2,3]. "Searching for the nature of stars with debris disks and planets". *Astronomy & Astrophysics*, 671, A136.

34. J. Blanco-Pozo, M. Perger, M. Damasso, G. Anglada Escudé, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. A long-period planet around GJ 1151 measured with CARMENES and HARPS-N data". *Astronomy & Astrophysics*, 671, A50.

33. I. Ribas, A. Reiners, M. Zechmeister, J. A. Caballero, et al. (incl. **C. Cifuentes**[1,2,3]). "The CARMENES search for exoplanets around M dwarfs. Guaranteed Time Observations Data Release 1 (2016-2020)". *Astronomy & Astrophysics*, 670, A139.

32. D. Kossakowski, M. Kürster, T. Trifonov, T. Henning, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs, Wolf 1069 b: Earth-mass planet in the habitable zone of a nearby, very low-mass star". *Astronomy & Astrophysics*, 670, A84.

31. M. Damasso, M. Perger, J. M. Almenara, D. Nardiello, et al. (incl. **C. Cifuentes**[1,2]). "A quarter century of spectroscopic monitoring of the nearby M dwarf Gl 514. A super-Earth on an eccentric orbit moving in and out of the habitable zone". *Astronomy & Astrophysics*, 666, A187.

30. P. Chaturvedi, P. bluhm, E. Nagel, A. P. Hatzes, et al. (incl. **C. Cifuentes**[1,2]). "TOI-1468: A system of two transiting planets, a super-Earth and a mini-Neptune, on opposite sides of the radius valley". *Astronomy & Astrophysics*, 666, A155.

29. R. Luque, G. Nowak, T. Hirano, D. Kossakowski, et al. (incl. **C. Cifuentes**[1,2]). "Precise mass determination for the keystone sub-Neptune planet transiting the mid-type M dwarf G 9-40". *Astronomy & Astrophysics*, 666, A154.

28. D. Kossakowski, M. Küster, Th. Henning, T. Trifonov, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. Stable radial-velocity variations at the rotation period of AD Leonis: A test case study of current limitations to treating stellar activity". *Astronomy & Astrophysics*, 666, A143.

27. J. A. Caballero, E. González-Alvarez, M. Brady, T. Trifonov, T. G. Ellis, C. Dorn, **C. Cifuentes**[1,2,4], et al. "A detailed analysis of the Gl 486 planetary system". *Astronomy & Astrophysics*, 665, A120.

26. S. V. Jeffers, J. R. Barnes, P. Schöfer, A. Quirrenbach, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs: Benchmarking the impact of activity in high-precision radial velocity measurements". *Astronomy & Astrophysics*, 663, A27.

25. J. Kemmer, S. Dreizler, D. Kossakowski, S. Stock, et al. (incl. **C. Cifuentes**[1,2]). "Discovery and mass measurement of the hot, transiting, Earth-sized planet, GJ 3929 b". *Astronomy & Astrophysics*, 659, A17.

24. N. Espinoza, E. Pallé, J. Kemmer, R. Luque, J. A. Caballero, **C. Cifuentes**[1,2], et al. "A Transiting, Temperate Mini-Neptune Orbiting the M Dwarf TOI-1759 Unveiled by TESS". *The Astronomical Journal*, 163, 133.

23. E. Marfil, H. M. Tabernero, D. Montes, J. A. Caballero, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. Stellar atmospheric parameters of target stars with SteParSyn". *Astronomy & Astrophysics*, 656, A162.

22. D. Kossakowski, J. Kemmer, P. Bluhm, S. Stock, J. A. Caballero, et al. (incl. **C. Cifuentes**[1,2]). "TOI-1201 b: A mini-Neptune transiting a bright and moderately young M dwarf". *Astronomy & Astrophysics*, 656, A124.

21. S. Roca-Fábrega, F. Llorente de Andrés, C. Chavero, **C. Cifuentes**[1,3], and R. de la Reza. "The bi-modal ^{7}Li distribution of the Milky Way's thin-disk dwarf stars. The role of Galactic-scale events and stellar evolution". *Astronomy & Astrophysics*, 656, A24.

20. F. Llorente de Andrés, C. Chavero, R. de la Reza, S. Roca-Fábrega, **C. Cifuentes**[1,3,4]. "The evolution of lithium in FGK dwarf stars: The Li rotation connection and the Li desert". *Astronomy & Astrophysics*, 654, A137.

19. V. Perdelwitz, M. Mittag, L. Tal-Or, J. H. M. M. Schmitt, et al. (incl. **C. Cifuentes**[1,3]). "CARMENES input catalog of M dwarfs VI. A time-resolved Ca II H&K catalog from archival data". *Astronomy & Astrophysics*, 652, A116.

18. P. J. Amado, F. F. Bauer, C. Rodríguez López, E. Rodríguez, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. Two terrestrial planets orbiting G 264-012 and one terrestrial planet orbiting Gl 393". *Astronomy & Astrophysics*, 650, A188.

17. T. Trifonov, J. A. Caballero, J. C. Morales, A. Seifahrt, I. Ribas, et al. (incl. **C. Cifuentes**[1,2]). "A nearby transiting rocky exoplanet that is suitable for atmospheric investigation". *Science*, 371, 6533, 1038-1041.

16. S. Dreizler, M. Crossfield, D. Kossakowski, P. Plavchan, S. Jeffers, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs–LP 714-47b (TOI 442.01): Populating the Neptune desert". *Astronomy & Astrophysics*, 644, A127.

15. J. Kemmer, S. Stock, D. Kossakowski, A. Kaminski, et al. (incl. **C. Cifuentes**[1,2]). "Discovery of a hot, transiting, Earth-sized planet and a second temperate, non-transiting planet around the M4 dwarf GJ 3473 (TOI-488)" . *Astronomy & Astrophysics*, 642, A236.

14. G. Nowak, R. Luque, H. Parviainen, E. Pallé, K. Molaverdikhani, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. Two planets on opposite sides of the radius gap transiting the nearby M dwarf LTT 3780". *Astronomy & Astrophysics*, 642, A173.

13. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, et al. "The CARMENES input catalogue of M dwarfs. V. Luminosities, colours, and spectral energy distributions". *Astronomy & Astrophysics*, 642, A115.

12. D. Baroch, J. C. Morales, I. Ribas, E. Herrero, A. Rosich, et al. (incl. **C. Cifuentes**[1]). "The CARMENES search for exoplanets around M dwarfs. Convective shift and starspot constraints from chromatic radial velocities". *Astronomy & Astrophysics* 641, A69.

11. P. Bluhm, R. Luque, N. Espinoza, E. Pallé, J. A. Caballero, et al (incl. **C. Cifuentes**[1,2]). "Precise mass and radius of a transiting super-Earth planet orbiting the M dwarf TOI-1235: a planet in the radius gap?". *Astronomy & Astrophysics* 639, A132.

10. E. Pallé, S. Nortmann, N. Casasayas-Barris, M. Lampón, et al. (incl. **C. Cifuentes**[1,2]). "A He I upper atmosphere around the warm Neptune in GJ 3470 b". *Astronomy & Astrophysics* 638, A61.

9. H. Martínez-Rodríguez, J. A. Caballero, **C. Cifuentes**[1,2,3], A. L. Piro, and R. Barnes. "Exomoons in the habitable zones of M dwarfs". *The Astrophysical Journal*, 887, 2, 261.

8. J. C. Morales, A. J. Mustill, I. Ribas, M. B. Davies, A. Reiners, et al. (incl. **C. Cifuentes**[1,2]). "A giant exoplanet around a very low mass star challenging formation models". *Science*, 365, 6460, 1441-1445.

7. M. Zechmeister, S. Dreizler, I. Ribas, A. Reiners, J. A. Caballero, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. Two temperate Earth-mass planet candidates around Teegardens star". *Astronomy & Astrophysics*. 627, A49.

6. A. Schweitzer, V. M. Passeger, **C. Cifuentes**[1,2,3], V. J. S. Béjar, et al. "The CARMENES search for exoplanets around M dwarfs. Radii and masses of the target stars". *Astronomy & Astrophysics* 625, A68.

5. R. González-Peinado, J. A. Caballero, D. Montes, and **C. Cifuentes**[1,3]. "Cool dwarfs in wide multiple systems: A curious quintuple system of a compact sun-like triple and a close M dwarf-white dwarf pair at a wide separation". *The Observatory*, Vol. 138, p. 292-298.

4. A. Kaminski, T. Trifonov, J. A. Caballero, A. Quirrenbach, I. Ribas, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. A Neptune-mass planet traversing the habitable zone around HD 180617". *Astronomy & Astrophysics*, 618, A115.

3. A. Reiners, M. Zechmeister, J. A. Caballero, I. Ribas, J. C. Morales, et al. (incl. **C. Cifuentes**[1,3]). "The CARMENES search for exoplanets around M dwarfs. High-resolution optical and near-infrared spectroscopy of 324 survey stars". *Astronomy & Astrophysics*, 612, A49.

2. T. Trifonov, M. Kürster, M. Zechmeister, L. Tal-Or, J. A. Caballero, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. First visual-channel radial-velocity measurements and orbital parameter updates of seven M-dwarf planetary systems". *Astronomy & Astrophysics*, 609, A117.

1. A. Reiners, I. Ribas, M. Zechmeister, J. A. Caballero, et al. (incl. **C. Cifuentes**[1,2]). "The CARMENES search for exoplanets around M dwarfs. HD 147379 b: A nearby Neptune in an early-M dwarf's temperate zone". *Astronomy & Astrophysics*, 609, A49.

B.3 Publications without peer review process

11. **C. Cifuentes**. "Astrophysical parameters of M dwarfs with exoplanets". PhD Thesis, Universidad Complutense de Madrid.

10. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, and the Carmenes Consortium. "Luminosities of cool stars". The Star-Planet Connection, Online Workshop.

9. Y. Shan, A. Reiners, P. J. Amado, V. J. S. Béjar, J. A. Caballero, **C. Cifuentes**[2], and the Carmenes Consortium. "CARMENES and the Frontiers of High-Resolution Spectroscopy for M dwarfs". *Plato Online Mission Conference 2021*.

8. **C. Cifuentes**, J. A. Caballero, and S. Agustí. "One Is the Loneliest Number: Multiplicity in Cool Dwarfs". *Research Notes of the AAS*, Vol. 5, Number 5.

7. E. Marfil, H. M. Tabernero, D. Montes, J. A. Caballero, F. J. Lázaro-Barrasa, et al. (incl. **C. Cifuentes**[2,3]). "Stellar atmospheric parameters of CARMENES GTO M dwarfs with spectral synthesis and SteParSyn". *The 20.5th Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun (CS20.5)*. Virtually anywhere.

6. A. Quirrenbach, and the CARMENES Consortium (incl. **C. Cifuentes**[2]). "The CARMENES M-dwarf planet survey". *Proceedings of the SPIE*, Vol. 11447, id. 114473C. Ground-based and Airborne Instrumentation for Astronomy. SPIE Astronomical Telescopes + Instrumentation 2020 Digital Forum. Online.

5. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, F. J. Abellán, et al. "Colours and luminosities of M dwarfs in the CARMENES input catalogue". *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*. Online.

4. M. Cortés-Contreras, A. J. Domínguez-Fernández, J. A. Caballero, et al. (incl. **C. Cifuentes**[2]). "Kinematics of M dwarfs in the CARMENES input catalogue". *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*. Online.

3. S. Roca-Fábrega, F. Llorente de Andrés, **C. Cifuentes**[1,3], C. Chavero, R. de la Reza and B. Montesinos. "The bimodal A(Li) distribution of Milky Way's thin disk stars and the Galactic scale events" *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*. Online.

2. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, A. Schweitzer, I. Ribas, P. J. Amado, and the CARMENES Consortium. "Spectral energy distributions and luminosities of M dwarfs in the CARMENES search for exoplanets". *Proceedings of the XIII Scientific Meeting of the Spanish Astronomical Society*. Highlights on Spanish Astrophysics X. Salamanca, Spain.

1. A. Quirrenbach, P. J. Amado, I. Ribas, A. Reiners, J. A. Caballero, et al. (incl. **C. Cifuentes**[2]). "CARMENES: high-resolution spectra and precise radial velocities in the red and infrared". *Proceedings of the SPIE*, Vol. 10702, id. 107020W. SPIE Astronomical Telescopes + Instrumentation. Austin, Texas, USA.

B.4 Conference Proceedings

15. M. Cortés-Contrera, J. A. Caballero, D. Montes, C. Cardona-Guillén, V. J. S. Béjar, **C. Cifuentes**, et al. "Kinematics, activity and youth in the CARMENES input catalogue of M dwarfs". *Proceedings of the XVI Scientific Meeting of the Spanish Astronomical Society* Highlights of Spanish Astrophysics XII. Granada, Spain

14. J. González-Payo, J. A. Caballero, J. Gorgas, M. Cortés-Contreras, M. C. Gálvez-Ortiz, and **C. Cifuentes** "Multiplicity of stars with planets". *Ibid.*

13. J. A. Caballero, **C. Cifuentes**, M. Cortés-Contreras, D. Montes, P. J. Amado, V. J. S. Béjar, et al. "Carmencita: Know thy star, know thy planet". *Ibid.*

12. J. A. Caballero, M. Capilla, M. A. Corbinos, I. Encinas, J. Miranzo, S. Turrado, and **C. Cifuentes**. "ARTof-COSMOS: aprendiendo a comunicar la astronomía al público". *Ibid.*

11. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, et al. "About the multiplicity of M dwarfs". *Proceedings of the XV Scientific Meeting of the Spanish Astronomical Society*, September 4–9, 2022.

10. **C. Cifuentes** and J. A. Caballero. "Music and astronomy. IV. The Astrophysical Brothers". *Ibid.*

9. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, and the Carmenes Consortium. "Luminosities of cool stars". *The Star-Planet Connection, On-line Workshop*, October 25–28, 2021.

8. Y. Shan, A. Reiners, P. J. Amado, V. J. S. Béjar, J. A. Caballero, **C. Cifuentes**, and the Carmenes Consortium. "CARMENES and the Frontiers of High-Resolution Spectroscopy for M dwarfs". *Plato Online Mission Conference 2021*, October, 2021.

7. E. Marfil, H. M. Tabernero, D. Montes, J. A. Caballero, F. J. Lázaro-Barrasa, et al. (incl. **C. Cifuentes**). "Stellar atmospheric parameters of CARMENES GTO M dwarfs with spectral synthesis and SteParSyn". *The 20.5th Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun (CS20.5)*. Virtually anywhere. March 2–4, 2021.

6. A. Quirrenbach, and the CARMENES Consortium (incl. **C. Cifuentes**). "The CARMENES M-dwarf planet survey". *Proceedings of the SPIE*, Vol. 11447, id. 114473C. Ground-based and Airborne Instrumentation for Astronomy. SPIE Astronomical Telescopes + Instrumentation 2020 Digital Forum. Online. December 14–18, 2020.

5. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, F. J. Abellán, et al. "Colours and luminosities of M dwarfs in the CARMENES input catalogue". *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*. Online. July 13–15 2020.

4. M. Cortés-Contreras, A. J. Domínguez-Fernández, J. A. Caballero, D. Montes, C. Cardona, V. J. S. Béjar, **C. Cifuentes**, et al. "Kinematics of M dwarfs in the CARMENES input catalogue". *Ibid.*

3. S. Roca-Fábrega, F. Llorente de Andrés, **C. Cifuentes**, C. Chavero, R. de la Reza and B. Montesinos. "The bimodal A(Li) distribution of Milky Way's thin disk stars and the Galactic scale events". *Ibid.*

2. **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, D. Montes, A. Schweitzer, I. Ribas, P. J. Amado, and the CARMENES Consortium. "Spectral energy distributions and luminosities of M dwarfs in the CARMENES search for exoplanets". *Proceedings of the XIII Scientific Meeting of the Spanish Astronomical Society*. Highlights on Spanish Astrophysics X. Salamanca, Spain. 16–20 July, 2018.

1. A. Quirrenbach, P. J. Amado, I. Ribas, A. Reiners, J. A. Caballero, et al. (incl. **C. Cifuentes**). "CARMENES: high-resolution spectra and precise radial velocities in the red and infrared". *Proceedings of the SPIE*, Vol. 10702, id. 107020W. SPIE Astronomical Telescopes + Instrumentation. Austin, Texas, USA. 10–15 June, 2018.

B.5 VizieR Online Data Catalogues

32. **CARMENES input catalogue of M dwarfs. IX** (**Cifuentes+**, 2025)
31. **Multiplicity of stars with planets** (González-Payo+, 2024)
30. **GJ 581 RVs and activity indices** (von Stauffenberg+, 2024)
29. **Stellar parameters of 286 CARMENES M dwarfs** (Mas-Buitrago+, 2024)
28. **HN Lib radial velocities** (González-Álvarez+, 2023)
27. **GJ 724 and GJ 3988 RV timeseries** (Gorrini+, 2023)
26. **Deep Transfer Learning of $T_{\rm eff}$ and [M/H]** (Bello-García+, 2023)
25. **Stars with debris disks and planets** (de la Reza+, 2023)
24. **GJ 1151 CARMENES and HARPS-N data** (Blanco-Pozo+, 2023)
23. **Wolf 1069 RV and stellar activity indices** (Kossakowski+, 2023)
22. **CARMENES search for exoplanets around M dwarfs** (Ribas+, 2023)
21. **AD Leo RV and stellar activity indices** (Kossakowski+, 2022)
20. **TOI-1468 photometry and radial velocities** (Chaturvedi+, 2022)
19. **VRI photometry and radial velocity of TOI-1759** (Espinoza+, 2022)
18. **Gl514 RVs and Activity diagnostics** (Damasso+, 2022)
17. **GJ 3929 b RVs and activity indicators** (Kemmer+, 2022)
16. **CARMENES stellar atmospheric parameters** (Marfil+, 2021)
15. **TOI-1201 RV and activity index** (Kossakowski+, 2021)
14. **Evolution of Li in FGK dwarf stars** (Llorente de Andrés+, 2021)
13. **CARMENES time-resolved CaII H&K catalog** (Perdelwitz+, 2021)
12. **G 264-012 and Gl 393 radial velocity curves** (Amado+, 2021)
11. **LP 714-47 (TOI 442) radial velocity curve** (Dreizler+, 2020)
10. **GJ 3473 (TOI-488) radial velocity curve** (Kemmer+, 2020)
9. **CARMENES input catalogue of M dwarfs. V** (**Cifuentes+**, 2020)
8. **Compilation of planets around M dwarfs** (Martínez-Rodríguez+, 2019)
7. **GJ 3512 radial velocity and light curves** (Morales+, 2019)
6. **Teegarden's Star RV and Hα curves** (Zechmeister+, 2019)
5. **Radii and masses of the CARMENES targets** (Schweitzer+, 2019)
4. **A Neptune-mass planet traversing the habitable zone around HD 180617** (Kaminski+, 2018)
3. **324 CARMENES M dwarfs velocities** (Reiners+, 2018)
2. **CARMENES radial velocity curves of 7 M-dwarf** (Trifonov+, 2018)
1. **HD147379 b velocity curve** (Reiners+, 2018)

B.6 Circulars

1. P. Bacci, M. Maestripieri, M. Facchini, M. D. Grazia, L. Tesi, et al. (incl. **C. Cifuentes**[6:Observer of ToO]). "MPEC 2020-A99: 2020 AV2". *Minor Planet Electronic Circulars*, No. 2020-A99. January 2020.

B.7 Book chapters and sections

1. *100 Conceptos de Astrobiología*, D. Barrado Navascués and S. Cabañero, and 50+ coauthors. Ed. INTA-CSIC (2022).

C Outreach

The symbols indicate oral (•) and poster (○) contributions.

- **Congresses and conferences:** •○ *Spanish Astronomical Society XV Meeting*, Tenerife, Spain (Sep 2022); ○ *Cool Stars 21*, Toulouse, France (Jun 2022); *The Star-Planet Connection*, ESO, *online* (Oct 2021); ○ *CAP 2021: Communicating Astronomy with the Public 2021*, IAU, *online* (May 2021); ○○ *Cool Stars 20.5 Virtual Meeting*, *online* (Mar 2021); ○○○ *Spanish Astronomical Society XIV.0 Meeting*, *online* (Jul 2020); ○*European Astronomical Society Annual Meeting*, *online* (Jul 2020); *1st meeting of the Spanish exoplanet network (Exonet)*, Granada, Spain (Feb 2019); • *Present and future science with CARMENES (RIA)*, Granada, Spain (Feb 2019); ○ *Encuentro RIA-SpaceTec: Instrumentación astronómica en España*, Madrid, Spain (Oct 2018); ○○○ *Spanish Astronomical Society XIII Meeting*, Salamanca, Spain (Jul 2018).
- **CARMENES consortium scientific meetings:** • 20th: *Naboo/online* (Nov 2024); • 16th: Göttingen, Germany (Jun 2022); 15th: *Endor/online* (Nov 2021); 14th: *Hoth/online* (May 2021); 13th: *Tatooine/online* (Nov 2020); 12th: *Arrakis/online* (Mar 2020); • 11th: Weimar, Germany (Nov 2019); • 10th: Seville, Spain (Nov 2019); • 9th: Barcelona, Spain (May 2019); 6th: Madrid, Spain (Apr 2017).
- **Schools, Workshops & Summer courses:** *Writing Better Research Papers and Proposals*, *online* (Nov 2023); *ESA's Madrid-Area Exoplanets Science Meeting (MAESM)*, Madrid, Spain (Oct 2022); *First Severo Ochoa IAA (SO-IAA) Workshop on scientific writing and oral presentations in astronomy*, Madrid, Spain (Sep 2020); *I Jornada de Invierno de Astrobiología*, Madrid, Spain (Jan 2020); *Machine learning and applications to physics (IPARCOS)*, Madrid, Spain (Dec 2019); *Cursos de Verano de El Escorial: 20 Años de Astrobiología en España*, Madrid, Spain (Jul 2019); *Gaia DR2 Exploration Lab*, Madrid, Spain (Jun 2018); *Virtual Observatory 11th School*, Madrid, Spain (May 2018); *Deep learning as a tool for scientific data analysis: Introduction and hands-on session*, Madrid, Spain (May 2018); *Aspectos metodológicos para el doctorado en el ámbito de las ciencias experimentales e ingenierías*, Madrid, Spain (Dec 2017); *ASTERICS VO School: Data Access, Discovery and Interoperability (DADI)*, Madrid, Spain (Nov 2017).
- **PhD-related events:** *II Jornadas para Doctorandos del CSIC*, *online* (Oct 2020); • *I Jornadas para Doctorandos del CSIC*, *online* (Jun 2019); • *Concurso "Tesis en 3 minutos"*, Madrid, Spain (Apr 2019); *Jornadas de Doctorandos UCM*, Madrid, Spain (Mar 2019); •○ *PhDay 2018*, Madrid, Spain (Nov 2018); *PhDay 2017*, Madrid, Spain (Dec 2017).
- **Other talks (journals):** • *Astroquery and its secrets (PyCoffee)*, Madrid, Spain (Oct, 2024).
- **Poster contributions as a co-author (without attendance):** ○○○ *Spanish Astronomical Society XVI Meeting*, Granada, Spain (Jul 2024); ○○ *European Astronomical Society Meeting 2022*, Valencia, Spain (Jul 2022); ○ *The Star-Planet Connection workshop*, *online* (Oct 2021); ○○○○ *TESS Science Conference 2 (TSC2)*, *online* (Aug 2021); ○○○ *European Astronomical Society Meeting 2020*, *online* (Jul 2020); ○○ *Annual Meeting of the Astronomische Gesellschaft 2017*, Göttingen, Germany (Sep 2017).

D Observational experience

D.1 As principal investigator

The number of on-site nights at the telescope is indicated in [brackets].

3. *Triple Trouble: Inner planets in hierarchical triple systems* (high-res spectroscopy).
   C. Cifuentes, O. Balsalobre-Ruza, J. Lillo-Box, J. González-Payo.
   2 nights with CARMENES @ CAHA-3.5 (approved for Semester 2025A)

2. *Identifying new M dwarfs spectroscopic binaries from Gaia RV data* (mid-res spectroscopy).
   C. Cifuentes, J. A. Caballero, E. Marfil, et al.
   8 nights with FIES @ NOT (86-NOT5/23A, 114-NOT16/22B) [2] and **5** service nights (fraction) with FIES @ NOT (114-NOT16/22B).

1. *Spectroscopy of poorly investigated late-type young stars in the sigma Orionis cluster* (long-slit spectroscopy).
   C. Cifuentes, J. A. Caballero, and A. De Burgos.
   Four half nights with ISIS Red+ @ WHT (164-WHT7/19B) [4]

D.2 As co-principal investigator

3. *Calibrating the abundances of M dwarfs with wide visual binaries* (high-res spectroscopy).
   C. Duque-Arribas, D. Montes, et al. (incl. **C. Cifuentes**).
   - **8** hours with SES @ STELLA (85-MULTIPLE-2/22B)
   - **3** nights with HERMES @ MERCATOR (85-MULTIPLE-2/22B)
   - **3.6** nights with CARMENES @ 3.5-m CAHA (23A-3.5-010)
   - **2.25** nights with CARMENES @ 3.5-m CAHA (23B-3.5-009)

2. *Formation origins of brown dwarfs from rotation rates* (image).
   N. Lodieu, et al. (incl. **C. Cifuentes**).
   5 nights with WFC @ INT (141-INT13/20B) [5]

1. *Optical spectroscopy of the only halo candidate in the CARMENES input catalogue* (long-slit spectroscopy).
   N. Lodieu, et al. (incl. **C. Cifuentes**).
   5 nights with OSIRIS @ GTC (142-GTC107/21A)

D.3 Other relevant collaborations

2. **One night** (remote) with ESPRESSO @ 8-m VLT (Period 107)
   M. R. Zapatero Osorio, E. González Álvarez, and **C. Cifuentes**.

1. **Two half nights** with CAFE @ 2.2-m CAHA (3–4 May 2017) [2]
   Open clusters in Cygnus Superbubble: Ber 87 & IC 4996 (image and long-slit spectroscopy).
   Part of the curricula in the MSc in Astrophysics.

E Academia

E.1 Degrees

- *Philosophiæ doctor* (PhD; Doctorado) – **Astrophysics** 2018–2023
  Universidad Autónoma de Madrid Madrid, Spain

  Thesis: "Astrophysical parameters of M dwarfs with exoplanets" • *Advisors:* Dr. José Antonio Caballero and Dr. Jorge Sanz Forcada • *Institutions:* Centro de Astrobiología (CAB, CSIC-INTA) and Max-Planck-Institut für Astronomie (MPIA)

- *Magister scientiæ* (MSc; Máster) – **Astrophysics** 2016–2017
  Universidad Complutense de Madrid Madrid, Spain

  Solar System and exoplanets • Cool stars and substellar objects • Stellar atmospheres • Interstellar medium • Galactic formation and evolution • Astronomical instrumentation • Experimental techniques in astrophysics • Data analysis and statistical techniques. (60 ECTS)

- *Magister scientiæ* (MSc; Máster) – **Renewable Energies** 2013–2014
  Universidad Autónoma de Madrid Madrid, Spain

  Wind and Solar power • Hydropower • Geothermal Energy • Biomass • Nuclear Energy: Fission and Fusion • Energy Storage: Hydrogen, Fuel Cells • Energy conversion, transportation and storage • Bioclimatic architecture • Environment, energy economy and sustainable development. (60 ECTS)

- *Baccalaureus scientiæ* (BSc; Licenciatura) – **Physics** 2007–2013
  Universidad de Salamanca Salamanca, Spain

  Specialization on Astrophysics and Earth Sciences (Climatology, Meteorology, Atmospheric thermodynamics and radiation). Photonics and electronics. (314 ECTS, Plan 1997)

E.2 Complementary courses

Programming, Data Science and OS

1. *Data Science: Productivity Tools* HarvardX, *online* (Dec 2018)
2. *Data Driven Astronomy with Python and SQL* University of Sidney, *online* (Dec 2017)
3. *Linux* Escuela Julián Besteiro, *online* (Mar 2011)
4. *Scientific Software in Mathematics: Mathematica and MATLAB* USAL, *Salamanca* (1–5 Jun 2009)

Astronomy and Astrophysics

1. *Cosmology* Australian National University, *online* (Mar 2016)
2. *Philosophy and the Sciences* University of Edinburgh, *online* (Nov 2015)
3. *The Science & Technology behind Astronomical Discovery* University of Edinburgh, *online* (Jul 2015)

Languages

1. *Inglés específico: presentaciones C1* Euroinnova, *online*, 30h (Oct–Nov 2020)
2. *English language immersion course* UIMP, Barcelona + Tenerife, 80h (Oct 2009 + Jun 2010)
3. *Upper-Intermediate Course in English* Servicio Central de Idiomas USAL, *Salamanca*, 60h (Sep 2011–Jan 2012)

Miscellanea

1. *Scopus* FECYT, *online*, 2h (Apr 18)
2. *Critical Thinking in Global Challenges* University of Edinburgh, *online* (Mar 2015)
3. *Wind Power: Fundamentals, technology and applications* CIEMAT, *online*, 50h (Dec 2014)
4. *Learning How to Learn* University of California, San Diego, *online* (Dec 2014)
5. *Fundamentals of Music Theory* University of Edinburgh, *online* (Sep 2014)

F Miscellanea

Volunteering

- Service at infopoints at the *2nd ESA Open Day*. 21 Oct 2023
- Reception, meet & greet & solar observation at the *1st ESA Open Day*. 26 Oct 2019
- General organization tasks at the *Spanish Astronomical Society XIII Meeting*. 16–20 Jul 2018
- Class representative in UCM Master's class (2016–2017).

Affiliations

- International Astronomical Union (IAU) (2024+).
- Red Española de Explotación Científica de *Gaia* (REG) (2022+).
- European Astronomical Society (EAS) (2021+).
- Sociedad Española de Astronomía (SEA) (2018+).
- Spanish Red Cross' Blood donor (2015+).

Additional experience

- Private classes (Mathematics, High school level).
- Telephone operator (*teleoperador*) in national and regional Spanish elections (2015-2016).

Other relevant information

- Professional qualifications: First Aids (July 2019).
- Driving License: B (2007+).

Astromusical projects

- Production of audiovisuals for two shows at *ESA Open Day 2023* (ESA) 21 Oct 2023
  J. A. Caballero & **C. Cifuentes** (*The Astrophysical Brothers*)
- Collaboration at *El Universo Cultural* (RNE) 21 Aug 2022
  Círculos Concéntricos, Radio Nacional de España
- Piano and bass guitar at *ESA Day 2022* (ESA) 22 Jun 2022
  J. A. Caballero & **C. Cifuentes**, et al. (*The Astrophysical Brothers & Friends*)
- *Astromúsica*'s playlist curation; support at *Diálogos en el Café Gijón* Mar–Apr 2022
  for *Cultura con C de Cosmos: Vida* (C^3)
- Piano, choirs and acting for *Féte de la Musique 2021* (ESA) 21 Jun 2021
  J. A. Caballero & **C. Cifuentes** (*The Astrophysical Brothers*)
- *Astromúsica*'s playlist curation for *Cultura con C de Cosmos* (C^3) Oct 2018–Mar 2019
- Other untitled one-man band projects (e.g. *Colma/Syzygy*) 2020+

These and other contributions can be found in `ccifuentesr.io/music` and `artofcosmos.com`.